

Backward intertwiners and a transitive commutant

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Abstract

We give a negative answer to the hyperinvariant-subspace problem by constructing, on every infinite-dimensional separable complex Hilbert space, a nonzero bounded norm-quasinilpotent operator with no nonzero proper closed hyperinvariant subspace. Its commutant is a proper strongly closed unital transitive complex operator algebra.

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1 Introduction

Let H be a complex Hilbert space and let $\mathcal{B}(H)$ denote its bounded linear operators. The *commutant* of $S \in \mathcal{B}(H)$ is

$$\{S\}' = \{A \in \mathcal{B}(H) : AS = SA\}.$$

A closed subspace is *hyperinvariant* for S if every operator in $\{S\}'$ leaves it invariant. An operator algebra is *transitive* if its only closed common invariant subspaces are $\{0\}$ and H . Thus S has no nonzero proper closed hyperinvariant subspace precisely when its commutant is transitive.

The hyperinvariant-subspace problem asks whether every bounded nonscalar operator on an infinite-dimensional separable complex Hilbert space has a nonzero proper closed hyperinvariant subspace. The construction below gives a negative answer. Its purpose is to exhibit the mechanism explicitly: the ordinary invariant subspaces of each weighted-shift fiber are coordinate tails, and a second, commuting operator moves those tails backward. Invariance under the commutant therefore imposes a strict descent that cannot persist on a probability space.

1.1 The construction theorem

Put $K = \ell^2(\mathbb{Z})$, with orthonormal basis $(e_j)_{j \in \mathbb{Z}}$, and define

$$K_{\geq h} = \overline{\text{span}}\{e_j : j \geq h\} \quad (h \in \mathbb{Z}), \quad K_{\geq -\infty} = K, \quad K_{\geq +\infty} = \{0\}.$$

Let $X = \mathbb{Z}_2$ be the compact additive group of 2-adic integers, the inverse limit of the groups $\mathbb{Z}/2^m\mathbb{Z}$. Equip it with completed Haar probability μ , put $\theta x = x + 1$, and set $H_0 = L^2(X, \mu; K)$. For $g \in L^\infty(X)$, the scalar multiplier is $(M_g f)(x) = g(x)f(x)$.

Theorem 1.1. *There are measurable positive weights $\beta_j : X \rightarrow (0, \infty)$, $j \in \mathbb{Z}$, and a contraction $V \in \mathcal{B}(H_0)$ with the following properties. The shifts*

$$S_x e_j = \beta_j(x) e_{j+1}, \quad (Sf)(x) = S_x f(x),$$

define a bounded operator S on H_0 , and:

1. For every $x \in X$, the closed invariant subspaces of S_x are exactly $K_{\geq h}$, $h \in \mathbb{Z} \cup \{-\infty, +\infty\}$.
2. The operator S is nonzero, $\|S\| = e^{-10}$, and for every integer $n \geq 1$,

$$\|S^n\| \leq \exp \left(-10 \left\lfloor \frac{(n+1)^2}{4} \right\rfloor \right). \quad (1)$$

3. The operator V commutes with S and has the form

$$(Vf)(\theta x) = V_x f(x), \quad V_x e_j = b_j(x) e_{j-1},$$

where $0 \leq b_j(x) \leq 1$ are measurable and are positive on a common conull set. On that set,

$$\overline{V_x K_{\geq h}} = K_{\geq h-1} \quad (h \in \mathbb{Z}), \quad \overline{V_x K} = K.$$

4. The unital complex algebra generated by S , V , and all scalar multipliers M_g is transitive.

All generators in the last assertion commute with S . The explicit power bound implies norm-quasinilpotence, $\lim_{n \rightarrow \infty} \|S^n\|^{1/n} = 0$, while the first assertion will give many ordinary invariant subspaces of the resulting operator.

Corollary 1.2. *Every infinite-dimensional separable complex Hilbert space admits a nonzero bounded norm-quasinilpotent operator with no nonzero proper closed hyperinvariant subspace. Its commutant is a proper strongly closed unital transitive complex operator algebra.*

Here the strong operator topology is pointwise norm convergence on H . Consequently the commutant in the corollary also gives a negative answer to the question whether every unital transitive operator algebra is strongly dense in $\mathcal{B}(H)$.

1.2 Context and relation to the companion

Douglas and Pearcy studied the connection between hyperinvariant subspaces and transitive algebras [4]. A major positive result is Arveson’s density theorem: a transitive algebra on a separable Hilbert space which contains a maximal abelian self-adjoint subalgebra of $\mathcal{B}(H)$ is strongly dense in $\mathcal{B}(H)$ [1, Theorem 3.3]. The scalar multiplier algebra used here has infinite multiplicity. Section 6 gives an explicit operator commuting with every such multiplier that is not itself a scalar multiplier, so the maximality hypothesis is absent.

Domar’s work on invariant subspaces of weighted translation spaces [3] and Grabiner’s work on unicellular shifts [6] provide the classical lattice ideas behind the fibers. Section 2 proves the precise quadratic weighted-translation criterion we need. Its convex-envelope and support-separation argument follows Domar’s method. The proof includes both normalized convolution-tail estimates; no general criterion for arbitrary weights is assumed. Our additional task is to link the fibers by a bounded commuting operator that moves their invariant tails backward. Opposite logarithmic changes at the two ends of the bilateral sequence make this possible, and one measurable normalization controls the operator over the whole base.

Atzmon and Sodin constructed completely indecomposable bilateral weighted shifts with spectrum the unit circle [2]; their 1999 paper left open whether those shifts have proper hyperinvariant subspaces. Here the extra commutant action is supplied by transport between different fibers. For a study of operators with shift-type invariant subspaces, see Kérchy [7]. The invariant- and hyperinvariant-subspace problems must remain distinct: the Kérchy–Pearcy preprint claiming an equivalence was withdrawn after a counterexample to its Theorem 3 [8]. The operator constructed here preserves every fixed tail $L^2(X; K_{\geq h})$; its hyperinvariant conclusion is proved directly.

The companion paper [9] constructs, for every prescribed irrational rotation, a continuous circle weight whose operator in the rotation factor has no nontrivial invariant projection in that factor. It derives the existence conclusion of Corollary 1.2 by the usual projection argument; see also Dykema [5, Proposition 2.1]. The present construction has a different purpose: it exhibits the ordinary invariant tails, the bounded operator that moves them backward, and the explicit estimate (1). The two proofs use different constructions. Our only proof imports from the companion are the two elementary facts stated below.

1.3 Proof overview and elementary inputs

The proof first makes the possible fiber subspaces completely explicit. Section 2 classifies invariant subspaces of a weighted translation whose logarithmic weights have opposite quadratic growth at the two ends. Section 3 constructs weights over $\mathbb{Z}_2$ satisfying that criterion in every fiber. Their decay also gives the uniform power estimate, which proves norm-quasinilpotence of the operator on H_0 rather than merely of its individual fibers.

To couple the fibers, we prescribe a map from e_j at x to a positive multiple of e_{j-1} at θx . Commutation with the forward shifts determines a recurrence for those multiples. Section 4 solves the recurrence in both directions. The logarithmic changes of the weights under θ have opposite signs at the two ends; the coefficients therefore decay in both directions. Dividing all coefficients in each fiber by their common supremum yields the contraction V without changing commutation.

Finally, a subspace invariant under every scalar multiplier has an orthogonal projection represented by a measurable projection field. Invariance under S makes each fiber a tail $K_{\geq s(x)}$. Section 5 uses the dense images under V_x to obtain

$$s(\theta x) \leq s(x) - 1 \quad \text{whenever } s(x) \in \mathbb{Z}.$$

Probability preservation excludes finite labels, and ergodicity excludes a nonconstant mixture of full and zero fibers. Section 6 then proves the corollary and records the structure of the example.

We use the following two elementary inputs, with the stated scope. Their complete proofs are in the companion [9].

1. *Fields from scalar multipliers* [9, Lemma 3.1]. For a probability space (Y, ν) , $K = \ell^2(\mathbb{Z})$, and $A \in \mathcal{B}(L^2(Y; K))$ commuting with every scalar multiplier, there is an essentially bounded field $A_y \in \mathcal{B}(K)$, unique almost everywhere, such that

$$(Af)(y) = A_y f(y), \quad \|A_y\| \leq \|A\| \quad \text{a.e.}$$

For every $v \in K$, the map $y \mapsto A_y v$ is measurable. Products and adjoints act fiberwise; an orthogonal projection has orthogonal projection fibers outside one null set.

2. *The commutant algebra* [9, Lemma 1.2]. For any bounded nonscalar operator on a complex Hilbert space, its commutant is a proper strongly closed unital complex algebra. If its only closed common invariant subspaces are zero and the whole space, that algebra is transitive.

The first input requires no base transformation, covariance, trace or factor assumption. We apply it only after proving reduction by all scalar multipliers. The second requires neither separability nor quasiniptence; the constructed operator will satisfy its nonscalarity hypothesis. The field input is used in Section 5, and the commutant input in Section 6.

2 The quadratic weighted-translation criterion

We first prove a criterion that makes every invariant fiber subspace a coordinate tail. The proof concerns one weighted sequence and does not involve the base transformation.

For a positive sequence $w = (w_j)_{j \in \mathbb{Z}}$, set

$$\ell^2(w) = \left\{ z : \sum_{j \in \mathbb{Z}} |w_j z_j|^2 < \infty \right\}, \quad \|z\|_{\ell^2(w)}^2 = \sum_{j \in \mathbb{Z}} |w_j z_j|^2.$$

If w is decreasing, right translation $(Rz)_j = z_{j-1}$ is a contraction on this space. For a sequence q_j on an integer interval, convexity means that $q_{j+1} - q_j$ is nondecreasing wherever the differences are defined; concavity means the reverse.

Lemma 2.1 (Weighted translation criterion). *Let $w = (w_j)_{j \in \mathbb{Z}}$ be positive and decreasing. Suppose that $\log w_j$ is convex for $j \leq 0$ and concave for $j \geq 0$, that*

$$\sum_{j \neq 0} w_j^{1/j} < \infty, \tag{2}$$

and that, for some $a > \log 3$,

$$\lim_{n \rightarrow \infty} \frac{\log w_n}{n^2} = -a, \quad \lim_{n \rightarrow \infty} \frac{\log w_{-n}}{n^2} = a. \tag{3}$$

Then the closed R -invariant subspaces of $\ell^2(w)$ are precisely

$$\{z : z_j = 0 \text{ for } j < h\}, \quad h \in \mathbb{Z} \cup \{-\infty, +\infty\}.$$

Consequently the shift $Te_j = (w_{j+1}/w_j)e_{j+1}$ on K has invariant lattice $\{K_{\geq h} : h \in \mathbb{Z} \cup \{-\infty, +\infty\}\}$.

The convex-envelope and support-separation argument below follows the method of Domar's weighted-translation work [3, Section 3]. We prove the stated quadratic specialization in full, including the normalized estimates for both convolution tails. The signed exponent for a negative index is essential: the corresponding summand is $w_{-n}^{-1/n}$ for $n > 0$.

Proof. Multiplying the weight by a constant, assume $w_0 = 1$. Put $P_n = \log w_{-n}$ and $Q_n = -\log w_n$ for $n \geq 0$. These sequences are convex, vanish at zero, and grow quadratically. For either sequence $G = P, Q$, convexity gives $G_n - G_{n-1} \geq G_n/n$. Put $\beta_j = w_{j+1}/w_j$. For $n \geq 1$,

$$\beta_{-n} = e^{-(P_n - P_{n-1})} \leq w_{-n}^{-1/n}, \quad \beta_{n-1} = e^{-(Q_n - Q_{n-1})} \leq w_n^{1/n},$$

The signed-root sum therefore gives

$$\beta_j \leq 1, \quad \sum_{j \in \mathbb{Z}} \beta_j < \infty. \quad (4)$$

In particular, R is bounded. If $|c_j|w_j \leq 1$ and $|b_j|/w_j \leq 1$, then for each integer $r \geq 1$,

$$\sum_j |b_j c_{j-r}| \leq \sum_j \frac{w_j}{w_{j-r}} \leq \sum_j \beta_{j-r} < \infty.$$

We prove the following support assertion: if nonzero b, c satisfy these bounds and

$$\sum_j b_j c_{j-r} = 0 \quad (r \geq 1), \quad (5)$$

then $\text{supp } b$ has a finite maximum, $\text{supp } c$ has a finite minimum, and

$$\max \text{supp } b \leq \min \text{supp } c. \quad (6)$$

Independent scaling of b, c gives the same assertion when the two weighted sequences are merely bounded. We will apply this assertion to a vector in an invariant subspace and to a continuous annihilator of that subspace. After proving support separation, the zero-shift pairing will identify the exact coordinate tail.

We first record an elementary fact about discrete convex envelopes. Let $f_0 = 0$ and $f_n \in [0, \infty]$ for $n \geq 1$, with f_n finite at arbitrarily large indices. Suppose there is a finite convex sequence $G \leq f$ with $G_0 = 0$ and $G_n \geq \delta n^2$ for all sufficiently large n , where $\delta > 0$. The pointwise supremum F of all finite convex minorants of f anchored at zero is finite: for any n , a finite obstacle at an index $k > n$ bounds every minorant at n by the chord from zero to f_k . The supremum is convex, since the defining midpoint inequality is preserved under pointwise suprema. It is therefore the greatest convex minorant, with $F_0 = 0$ and $F \geq G$. If $n \geq 1$ and $F_{n+1} - 2F_n + F_{n-1} > 0$, then

$$F_n = f_n < \infty. \quad (7)$$

Indeed, otherwise raising F_n slightly preserves convexity at n , strengthens it at neighboring indices, and stays below f , contradicting maximality. Also the first differences of F tend to infinity, so there are arbitrarily large indices satisfying this strict inequality. We call them contact indices.

First suppose $h = \min \text{supp } c$ exists. If $\text{supp } b$ were unbounded above, take the greatest convex minorant B of $f_0 = 0$, $f_n = -\log |b_n|$ for $n \geq 1$, where $-\log 0 = \infty$. The obstacle at zero is an artificial anchor; the actual coefficient only needs $|b_0| \leq 1$, which follows from normalization. The

sequence Q is a minorant, so $B \geq Q$. Writing $v_n = e^{-B_n}$, we have $|b_n| \leq v_n \leq w_n$ for $n \geq 0$, and $|b_p| = v_p$ at arbitrarily large contact indices p . For such $p > h$, the correlation at $r = p - h$ gives

$$|c_h| \leq \sum_{k \geq 1} \frac{v_{p+k}}{v_p} |c_{h+k}|. \quad (8)$$

Fix an integer $p_0 \geq \max(1, h + 1)$. For $p \geq p_0$, convexity of B and (4) give

$$\frac{v_{p+k}}{v_p} |c_{h+k}| \leq \frac{v_{p_0+k}}{v_{p_0} w_{h+k}} \leq \frac{w_{p_0+k}}{v_{p_0} w_{h+k}} \leq \frac{\beta_{h+k}}{v_{p_0}}.$$

This is summable in k . For each fixed $k \geq 1$ the left side tends to zero as $p \rightarrow \infty$, because the first differences of B tend to infinity. Dominated convergence contradicts (8). Hence $t = \max \text{supp } b$ exists. If $t > h$, the correlation with $r = t - h$ consists of the single nonzero term $b_t c_h$, a contradiction. Thus $t \leq h$.

The corresponding argument when b is known to be bounded above is obtained by applying this case to

$$\tilde{w}_j = 1/w_{-j}, \quad \tilde{c}_j = b_{-j}, \quad \tilde{b}_j = c_{-j}.$$

These weights satisfy the same hypotheses, and their correlations are (5) after reindexing. Thus it remains only to exclude the case in which $\text{supp } c$ is unbounded below and $\text{supp } b$ is unbounded above.

In that case let A_m be the convex envelope of $-\log |c_{-m}|$ for $m \geq 1$, anchored at zero, and let B_p be the convex envelope of $-\log |b_p|$ for $p \geq 1$, also anchored at zero. Both envelopes have value zero at the anchor, while the actual coefficients satisfy only $|c_0|, |b_0| \leq 1$. Their convex minorants P, Q give $A \geq P$, $B \geq Q$. Define

$$u_{-m} = e^{A_m}, \quad u_p = e^{-B_p} \quad (m, p \geq 0).$$

Since $A_0 = B_0 = 0$ and $A_1 \geq P_1 \geq 0$, $B_1 \geq Q_1 \geq 0$, convexity makes all first differences of A and B nonnegative. The sequence u is therefore positive and decreasing, with $u_0 = 1$; its logarithm is convex on the negative half-line and concave on the positive half-line. The obstacle inequalities and $u_j \geq w_j$ for $j \leq 0$, $u_j \leq w_j$ for $j \geq 0$ give

$$|c_j| \leq u_j^{-1}, \quad |b_j| \leq u_j \quad (j \in \mathbb{Z}). \quad (9)$$

The quadratic lower bounds for A, B also give

$$\sum_{j \in \mathbb{Z}} \frac{u_{j+1}}{u_j} < \infty. \quad (10)$$

Choose $\log 3 < \lambda < a$. There are arbitrarily large m with

$$A_{m+1} - 2A_m + A_{m-1} > 2\lambda. \quad (11)$$

Otherwise double summation gives $A_m \leq \lambda m^2 + O(m)$, contrary to $A_m \geq P_m$ and $P_m/m^2 \rightarrow a$. Set $\alpha_m = A_m - A_{m-1}$ and $\gamma_p = B_p - B_{p-1}$. Convexity and the quadratic lower bounds imply that both sequences are nondecreasing and tend to infinity. We match these slopes to choose two contact coefficients whose product dominates the other terms in the middle of one correlation. The two remaining tails will then be negligible relative to that product. For each sufficiently large such m , choose the last index $p \geq 1$ with

$$\gamma_p \leq \alpha_m + \lambda < \gamma_{p+1}. \quad (12)$$

The choice exists because the nondecreasing sequence γ_p tends to infinity; moreover $p \rightarrow \infty$ with m . Equations (11)–(12) yield

$$\gamma_{p+1} > \alpha_m + \lambda, \quad \alpha_{m+1} > \gamma_p + \lambda.$$

Both m and p are contact indices. Therefore

$$|b_p c_{-m}| = u_p / u_{-m} =: M > 0. \quad (13)$$

The sequence

$$F(n) = \log \frac{u_{p+n}}{u_{-m+n}} = -B_{p+n} - A_{m-n}, \quad -p \leq n \leq m,$$

is concave. The preceding slope inequalities give $F(1) - F(0) < -\lambda$ and $F(-1) - F(0) < -\lambda$. Consequently

$$\frac{u_{p+n}}{u_{-m+n}} \leq M e^{-\lambda|n|} \quad (-p \leq n \leq m). \quad (14)$$

Put $q = m + p$ and use the correlation for $r = q$, with $j = p + n$. After isolating the term $n = 0$, the bounds (9)–(13) give

$$M \leq \sum_{n \neq 0} \frac{u_{p+n}}{u_{-m+n}}.$$

The part with $-p \leq n \leq m$ is at most

$$MC_\lambda, \quad C_\lambda = 2 \sum_{n \geq 1} e^{-\lambda n} = \frac{2}{e^\lambda - 1} < 1.$$

At $n = m$, (14) gives $u_q \leq M$. Thus the part with $n > m$, divided by M , is at most

$$\sum_{k \geq 1} \frac{u_{k+q}}{u_k u_q}.$$

Concavity of $\log u$ on the positive half-line implies, for $q \geq 1$,

$$\frac{u_{k+q}}{u_k u_q} \leq \frac{u_{k+1}}{u_1 u_k}.$$

The majorant is summable by (10); for each fixed $k \geq 1$, the term on the left tends to zero as $q \rightarrow \infty$, because u_{q+k}/u_q is a product of k step ratios tending to zero. Dominated convergence shows that this tail is $o(M)$. For the other tail, put $v_k = 1/u_{-k}$. Then $\log v$ is concave, its step ratios are summable, and (14) at $n = -p$ gives $v_q \leq M$. Writing $k = -p - n$ bounds this tail, divided by M , by

$$\sum_{k \geq 1} \frac{v_{k+q}}{v_k v_q} = o(1)$$

by the same argument. We obtain $1 \leq C_\lambda + o(1)$, a contradiction. This proves (6).

Finally let L be a nonzero proper closed R -invariant subspace. Its continuous annihilators have the form $c \mapsto \sum_j b_j c_j$ with $b/w \in \ell^2$. For $c \in L$ and such an annihilator b , the correlations vanish for all $r \geq 0$, and the two required weighted boundedness conditions hold. Fix a nonzero annihilator b^* and put $t = \max \text{supp } b^*$. Support separation shows that every nonzero $c \in L$ is supported at indices at least t ; orthogonality at $r = 0$ then forces $c_t = 0$. Since $L \neq \{0\}$, the union of supports of its elements is a nonempty subset of $\{t+1, t+2, \dots\}$ and therefore has a least index h . Choose $c^* \in L$ with $c_h^* \neq 0$. Every $c \in L$ vanishes below h . Conversely, support separation with c^* shows that every nonzero annihilator b is supported at indices at most h , and orthogonality at $r = 0$ forces

$b_h = 0$. All annihilators therefore vanish on the coordinate tail beginning at h . Hahn–Banach and closedness imply that this tail is contained in L . Thus L is exactly that tail.

Finally the map $c \mapsto (w_j c_j)_j$ is a unitary from $\ell^2(w)$ onto K . It carries R to the bounded weighted shift $Te_j = (w_{j+1}/w_j)e_{j+1}$ and preserves every coordinate-vanishing condition. This proves the last assertion as well. $\square$

3 The 2-adic weights

We now construct the measurable family of shifts. Their logarithmic weights must satisfy the criterion of Section 2 while changing in opposite directions at the two ends when the base moves by one.

3.1 The base transformation

For the inverse-limit base $X = \mathbb{Z}_2$ fixed in the introduction and $m \geq 1$, let

$$N_m = 2^m, \quad T_m(x) = x \bmod N_m \in \{0, \dots, N_m - 1\}.$$

Thus each residue is uniformly distributed. The map $\theta x = x + 1$ is an invertible probability-preserving transformation, and

$$T_{m+1}(x) - T_m(x) \in \{0, N_m\}, \quad (15)$$

$$T_m(\theta x) - T_m(x) = \begin{cases} 1, & T_m(x) < N_m - 1, \\ -(N_m - 1), & T_m(x) = N_m - 1. \end{cases} \quad (16)$$

The finite residue classes generate the Borel sigma-algebra of X .

Lemma 3.1. *The transformation θ is ergodic.*

Proof. If E is invariant modulo null sets, θ cyclically permutes the N_m residue classes, so

$$\mu(E \cap \{T_m = r\}) = \frac{\mu(E)}{N_m} \quad (0 \leq r < N_m).$$

Thus E is independent of every finite union of residue classes. By the Monotone Class Theorem it is independent of every measurable set, in particular itself. Hence $\mu(E) = \mu(E)^2$. $\square$

3.2 The weights

Fix

$$c = 10, \quad D = 100, \quad \ell_0 = 1, \quad \ell_m = 8^m \ (m \geq 1).$$

The integer blocks $I_m = \{k : \ell_{m-1} \leq k < \ell_m\}$ partition the positive integers. Set $J_1 = 0$ and $J_{m+1} = J_m + DN_m$; equivalently $J_m = D(N_m - 2)$. Define $a_0^+(x) = a_0^-(x) = c$, and for $k \in I_m$ put

$$a_k^+(x) = c(k + 1) + J_m + DT_m(x), \quad (17)$$

$$a_k^-(x) = c(k + 1) + J_m + D(N_m - 1 - T_m(x)). \quad (18)$$

On the full integer axis define

$$a_j(x) = \begin{cases} a_j^+(x), & j \geq 0, \\ a_{-j-1}^-(x), & j < 0, \end{cases} \quad \beta_j(x) = e^{-a_j(x)}. \quad (19)$$

The two central weights satisfy $\beta_{-1}(x) = \beta_0(x) = e^{-c}$ for every x . The linear terms in $a_k^\pm$ give quadratic growth of the cumulative logarithms. The smaller residue corrections change by $+D$ and $-D$ under θ unless the residue wraps around; the increments of J_m preserve monotonicity across block boundaries. We first check the invariant-subspace criterion, then use those opposite changes to construct the backward intertwiner.

Proposition 3.2. *For every $x \in X$, the operator $S_x e_j = \beta_j(x) e_{j+1}$ is bounded, has norm e^{-c} , and has precisely the standard tails as its closed invariant subspaces. For each $v \in K$, the map $x \mapsto S_x v$ is measurable.*

Proof. We verify positivity, regularity, and growth separately. The corrections to $c(k+1)$ in (17)–(18) are nonnegative. Thus

$$0 < \beta_j(x) \leq e^{-c}, \quad \sum_{j \in \mathbb{Z}} \beta_j(x) \leq 2 \sum_{k \geq 0} e^{-c(k+1)} < \infty. \quad (20)$$

Within a block, both $a_k^\pm$ increase by c at each step. At a boundary from I_m to I_{m+1} , put $\delta = T_{m+1}(x) - T_m(x) \in \{0, N_m\}$. The increments are, respectively,

$$c + DN_m + D\delta \geq c, \quad c + D(N_{m+1} - \delta) \geq c.$$

Also $a_1^\pm \geq 2c > a_0^\pm$. Consequently β_j is nonincreasing for $j \geq 0$ and nondecreasing for $j \leq 0$, including the equality at $-1, 0$.

Set $w_0(x) = 1$ and define $w_{j+1}(x)/w_j(x) = \beta_j(x)$ for all $j \in \mathbb{Z}$. This gives a positive decreasing sequence. Since $\log w_{j+1} - \log w_j = \log \beta_j$, its logarithm has the required convexity and concavity.

For $k \in I_m$ we have $N_m \leq 2k^{1/3}$, and hence

$$0 \leq a_k^\pm(x) - c(k+1) \leq 4Dk^{1/3} \quad (k \geq 1). \quad (21)$$

The bound is uniform in x . Summing gives, as $n \rightarrow \infty$,

$$\log w_n(x) = - \sum_{k=0}^{n-1} a_k^+(x) = - \frac{cn(n+1)}{2} + O(n^{4/3}), \quad (22)$$

$$\log w_{-n}(x) = \sum_{k=0}^{n-1} a_k^-(x) = \frac{cn(n+1)}{2} + O(n^{4/3}). \quad (23)$$

Thus (3) holds with $a = c/2 = 5 > \log 3$. Moreover the lower bounds $a_k^\pm \geq c(k+1)$ imply

$$w_n(x)^{1/n} \leq e^{-c(n+1)/2}, \quad w_{-n}(x)^{-1/n} \leq e^{-c(n+1)/2} \quad (n \geq 1).$$

These bounds prove (2). Lemma 2.1 therefore applies to every fiber.

The norm statement follows from (20) and $\beta_0 = e^{-c}$. Each coefficient β_j is measurable, so $x \mapsto S_x e_j$ is measurable. Finite basis expansions and the uniform norm bound give strong measurability on every vector of K . $\square$

We may now define $S \in \mathcal{B}(H_0)$ by $(Sf)(x) = S_x f(x)$. It has norm e^{-c} . For each $g \in L^\infty(X)$, the scalar multiplier $(M_g f)(x) = g(x)f(x)$ commutes with S . Before constructing the operator that links different fibers and moves their tails backward, we record the uniform power bound for S .

Proposition 3.3. *The operator S is nonzero and norm-quasinilpotent. More precisely, for all integers $n \geq 1$,*

$$\|S^n\| \leq \exp \left(-c \left\lfloor \frac{(n+1)^2}{4} \right\rfloor \right).$$

Proof. Define $d(j) = j + 1$ for $j \geq 0$ and $d(j) = -j$ for $j < 0$. The weight definitions imply $a_j(x) \geq cd(j)$ for every x, j . Since

$$S_x^n e_j = \exp \left(- \sum_{r=0}^{n-1} a_{j+r}(x) \right) e_{j+n},$$

the norm of each fiber power is bounded by the claimed quantity. Indeed the minimum of $\sum_{r=0}^{n-1} d(j+r)$ over $j \in \mathbb{Z}$ is attained by an interval centered as closely as possible on the two minima at -1 and 0 . For $n = 2m$ this minimum is $m(m+1)$; for $n = 2m+1$ it is $(m+1)^2$. Both values are $\lfloor (n+1)^2/4 \rfloor$. The bound is independent of x , so integrating the fiber estimate proves it for S^n itself. Consequently $\lim_{n \rightarrow \infty} \|S^n\|^{1/n} = 0$. Finally $S(1 \otimes e_0) = e^{-c}(1 \otimes e_1) \neq 0$. $\square$

4 A bounded backward intertwiner

We seek an operator which sends e_j in the fiber at x to a positive multiple of e_{j-1} in the fiber at θx . Commutation with the forward shift determines a recurrence for those multiples. The opposite logarithmic drifts at the two ends of the fiber make its solutions bounded after normalization.

4.1 Two-ended drift

Write $\Delta a_j(x) = a_j(\theta x) - a_j(x)$. A block I_m has an exceptional residue at x when $T_m(x) = N_m - 1$. Since

$$\sum_{m \geq 1} \mu\{T_m = N_m - 1\} = \sum_{m \geq 1} 2^{-m} < \infty,$$

the Borel–Cantelli Lemma shows that almost every x has only finitely many such blocks. Intersect this conull set over all iterates of θ to obtain a measurable θ -invariant conull set X_* . No independence of the exceptional events is needed.

Lemma 4.1. *For every $x \in X_*$ there is a finite $C(x)$ such that, for all $n, q \geq 1$,*

$$\sum_{k=0}^{n-1} \Delta a_k(x) \geq Dn - C(x), \tag{24}$$

$$\sum_{k=-q}^{-1} \Delta a_k(x) \leq -Dq + C(x). \tag{25}$$

Proof. For $k \in I_m$, Equation (16) gives

$$\Delta a_k^+(x) = D - DN_m \mathbf{1}_{\{T_m = N_m - 1\}}(x), \quad \Delta a_k^-(x) = -\Delta a_k^+(x).$$

Both central increments $\Delta a_0^\pm$ vanish. Thus the first sum is

$$D(n-1) - D \sum_{\substack{m \geq 1 \\ T_m(x) = N_m - 1}} N_m |I_m \cap [1, n]|.$$

Under $k \mapsto -k-1$, the negative-index sum is the negative of the same expression with $n = q$. Both claims follow on taking

$$C(x) = D + D \sum_{\substack{m \geq 1 \\ T_m(x) = N_m - 1}} N_m |I_m| < \infty.$$

$\square$

The constant $C(x)$ may be unbounded and nonintegrable as a function of x . Only its pointwise finiteness is used below.

4.2 Solving and normalizing the recurrence

For positive coefficients $b_j(x)$, sending e_j at x to $b_j(x)e_{j-1}$ at θx would commute with S precisely when

$$b_j(x)\beta_{j-1}(\theta x) = \beta_j(x)b_{j+1}(x).$$

Figure 1 shows the two paths in this coefficient identity. The left side is obtained by moving backward and then forward, and the right side in the opposite order.

$$\begin{array}{ccc}
 e_j \text{ at } x & \xrightarrow{S_x : \beta_j(x)} & e_{j+1} \text{ at } x \\
 \downarrow V_x : b_j(x) & & \downarrow V_x : b_{j+1}(x) \\
 e_{j-1} \text{ at } \theta x & \xrightarrow{S_{\theta x} : \beta_{j-1}(\theta x)} & e_j \text{ at } \theta x
 \end{array}$$

Figure 1: The coefficient identity at a fixed input fiber. Both paths take e_j at x to a multiple of e_j at θx . The horizontal maps move forward in the coordinate, while the vertical maps move backward and translate the base.

We first solve this equation with initial value one, then divide all coefficients in each fiber by a common bound. For every $x \in X$ set $\tilde{b}_0(x) = 1$ and define positive coefficients at all integer indices by

$$\tilde{b}_{j+1}(x) = \tilde{b}_j(x) \frac{\beta_{j-1}(\theta x)}{\beta_j(x)} \quad (j \in \mathbb{Z}). \quad (26)$$

Every coefficient is a finite product of positive measurable functions and their reciprocals, so it is finite and measurable.

Lemma 4.2. *For $x \in X_*$, the coefficients $\tilde{b}_j(x)$ tend to zero as $j \rightarrow +\infty$ and as $j \rightarrow -\infty$. Consequently there are measurable functions $b_j : X \rightarrow [0, 1]$ which are positive on X_* and satisfy*

$$b_{j+1}(x)\beta_j(x) = b_j(x)\beta_{j-1}(\theta x) \quad (x \in X, j \in \mathbb{Z}). \quad (27)$$

Proof. Taking logarithms in (26) and summing forward gives

$$\begin{aligned}
 \log \tilde{b}_n(x) &= \sum_{k=0}^{n-1} (a_k(x) - a_{k-1}(\theta x)) \\
 &= - \sum_{k=0}^{n-1} \Delta a_k(x) + a_{n-1}(\theta x) - a_{-1}(\theta x), \quad n \geq 1.
 \end{aligned} \quad (28)$$

Summing backward from index 0 gives the different sign

$$\begin{aligned}
 \log \tilde{b}_{-q}(x) &= - \sum_{k=-q}^{-1} (a_k(x) - a_{k-1}(\theta x)) \\
 &= \sum_{k=-q}^{-1} \Delta a_k(x) + a_{-q-1}(\theta x) - a_{-1}(\theta x), \quad q \geq 1.
 \end{aligned} \quad (29)$$

Here $a_{-1} = c$. The boundary indices have different forms: $a_{n-1} = a_{n-1}^+$, whereas $a_{-q-1} = a_q^-$. Using $a_0^\pm = c$ and Equation (21), we obtain

$$a_{n-1}(\theta x) - c \leq c(n-1) + 4D(n-1)^{1/3}, \quad a_{-q-1}(\theta x) - c \leq cq + 4Dq^{1/3}.$$

Combining these bounds with Lemma 4.1, we obtain the explicit estimates

$$\begin{aligned} \log \tilde{b}_n(x) &\leq -(D-c)n + C(x) - c + 4D(n-1)^{1/3}, \\ \log \tilde{b}_{-q}(x) &\leq -(D-c)q + C(x) + 4Dq^{1/3}. \end{aligned}$$

Since $D-c = 90 > 0$, both right sides tend to $-\infty$. These estimates are pointwise in x ; they do not require an integrable bound on $C(x)$.

The countable supremum

$$L(x) = \sup_{j \in \mathbb{Z}} \tilde{b}_j(x)$$

is measurable, and $1 \leq L(x) < \infty$ on X_* . Define

$$b_j(x) = \begin{cases} \tilde{b}_j(x)/L(x), & x \in X_*, \\ 0, & x \notin X_*. \end{cases}$$

All indices at a fixed x are divided by the same number, so (26) yields (27). Off X_* both sides of that identity vanish. $\square$

Define a backward shift $V_x : K \rightarrow K$ by $V_x e_j = b_j(x) e_{j-1}$. Every V_x has norm at most one, and $x \mapsto V_x v$ is measurable for every $v \in K$ by finite-coordinate approximation. It defines $V \in \mathcal{B}(H_0)$ by

$$(Vf)(\theta x) = V_x f(x), \quad \text{equivalently } (Vf)(\theta x, j-1) = b_j(x) f(x, j). \quad (30)$$

Invariance of μ gives $\|Vf\|^2 = \int_X \|V_x f(x)\|^2 d\mu(x) \leq \|f\|^2$.

Proposition 4.3. *The contraction V commutes with S . For every $x \in X_*$ and $h \in \mathbb{Z}$,*

$$\overline{V_x K_{\geq h}} = K_{\geq h-1}, \quad \overline{V_x K} = K.$$

Proof. At the same output base point θx , the two compositions have coefficients

$$S_{\theta x} V_x e_j = b_j(x) \beta_{j-1}(\theta x) e_j, \quad V_x S_x e_j = \beta_j(x) b_{j+1}(x) e_j.$$

They agree by (27), first on the basis and then on K . Equation (30) now gives $SV = VS$. On X_* every $b_j(x) > 0$. The image of a tail therefore contains every basis vector in the preceding tail, up to a nonzero scalar, and is contained in that preceding tail. This proves the density assertions. $\square$

The density conclusion, rather than surjectivity, is exactly what is needed for closed invariant subspaces. No bounded inverse of V_x or V is asserted.

5 All common invariant subspaces

We now show that the scalar multipliers, S , and V have no proper nonzero closed common invariant subspace. The first step accounts for every closed subspace invariant under the multipliers, without assuming in advance that it is defined fiber by fiber.

5.1 Multipliers force a measurable field of projections

Let $M \subseteq H_0$ be a closed subspace invariant under S , V , and every scalar multiplier. Since $M_{\bar{g}} = M_g^*$ belongs to the same multiplier family, M reduces each multiplier. Its orthogonal projection P therefore commutes with all of them. The field lemma [9, Lemma 3.1], applied with the probability space (X, μ) and $K = \ell^2(\mathbb{Z})$, gives

$$(Pf)(x) = P_x f(x), \quad M = \{f \in H_0 : f(x) \in \text{ran } P_x \text{ for almost every } x\},$$

where the P_x are orthogonal projections outside one null set and $x \mapsto P_x v$ is measurable for each $v \in K$. These are exactly the projection and measurability conclusions needed below; no covariance property is imposed on this field.

5.2 Fiber invariance and measurable tail labels

For this fixed subspace M and its projection field P_x , the global invariance identities $(I - P)SP = 0$ and $(I - P)VP = 0$ imply

$$(I - P_x)S_x P_x = 0, \tag{31}$$

$$(I - P_{\theta x})V_x P_x = 0 \tag{32}$$

almost everywhere. To justify the passage, apply each global identity to the constant field e_j . In the second identity evaluate the output at θx and use probability preservation. The resulting equalities hold for every j on a common conull set and extend to all of K by boundedness. Intersect this set, the projection set, and X_* over all integer iterates of θ . We can henceforth use all these identities on a single invariant conull set. This set is chosen after fixing M ; no intersection over all candidate subspaces is required.

By Proposition 3.2 and (31), there is a label

$$s(x) \in \mathbb{Z} \cup \{-\infty, +\infty\}, \quad \text{ran } P_x = K_{\geq s(x)}.$$

It is measurable: for every $j \in \mathbb{Z}$,

$$\{x : s(x) \leq j\} = \{x : \langle P_x e_j, e_j \rangle = 1\} \tag{33}$$

on the conull set, and the label can be set to $+\infty$ elsewhere. Equation (32) and Proposition 4.3 give two implications:

$$s(x) = -\infty \implies s(\theta x) = -\infty, \tag{34}$$

$$s(x) \in \mathbb{Z} \implies s(\theta x) \leq s(x) - 1. \tag{35}$$

The first uses $\overline{V_x K} = K$. The second uses $\overline{V_x K_{\geq h}} = K_{\geq h-1}$ and the inclusion $K_{\geq h-1} \subseteq \text{ran } P_{\theta x}$. When the input fiber is zero, there is no restriction on the output label.

5.3 Strict descent on a probability space

The final measure argument is useful independently of the construction.

Lemma 5.1. *Let θ be an invertible ergodic probability-preserving transformation of (Y, ν) . Suppose a measurable function $s : Y \rightarrow \mathbb{Z} \cup \{-\infty, +\infty\}$ satisfies, almost everywhere,*

$$s(x) = -\infty \implies s(\theta x) = -\infty, \quad s(x) \in \mathbb{Z} \implies s(\theta x) \leq s(x) - 1.$$

Then $s = -\infty$ almost everywhere or $s = +\infty$ almost everywhere.

Proof. All inclusions in this proof are modulo null sets. Put $E = \{s = -\infty\}$. The first implication gives $\theta E \subseteq E$. Probability preservation makes this inclusion an equality modulo null sets, so ergodicity gives $\nu(E) \in \{0, 1\}$. The case $\nu(E) = 1$ is one conclusion.

Suppose $\nu(E) = 0$. For each integer j , put $F_j = \{s \leq j\}$. Both implications give

$$\theta F_j \subseteq F_{j-1} \subseteq F_j.$$

It follows that $\nu(F_{j-1}) = \nu(F_j)$. For any fixed j , continuity from above now gives

$$\nu(F_j) = \nu\left(\bigcap_{r \geq 0} F_{j-r}\right) = \nu(E) = 0.$$

The countable union of the F_j is $\{s < +\infty\}$, proving the other conclusion. $\square$

Applying Lemma 5.1 shows that $P_x = I$ almost everywhere or $P_x = 0$ almost everywhere. We have therefore proved that the unital complex algebra generated by S , V , and all scalar multipliers is transitive.

6 The commutant and structural consequences

Proof of Theorem 1.1 and Corollary 1.2. Proposition 3.2 proves the fiber classification and norm statement in Theorem 1.1, and Proposition 3.3 gives the uniform power bound and nonzeroness. Proposition 4.3 constructs the commuting contraction with its dense tail images. Section 5 proves transitivity of the algebra generated by S , V , and the scalar multipliers. These establish all assertions of the theorem.

That transitive algebra is contained in $\{S\}'$, so any closed $\{S\}'$ -invariant subspace is zero or all of H_0 . The nonzero norm-quasinilpotent operator S is nonscalar. The commutant lemma [9, Lemma 1.2], with this S on the complex Hilbert space H_0 , now implies that $\{S\}'$ is a proper strongly closed unital complex algebra. Together with the lattice conclusion, it is transitive.

Finally, the finite residue classes form a countable generating algebra on X , so $L^2(X)$ and H_0 are separable. The space H_0 is infinite-dimensional. Every infinite-dimensional separable complex Hilbert space is unitarily isomorphic to H_0 ; conjugation by such a unitary transfers S , its norm estimates, its commutant, and the invariant-subspace conclusions. This proves the corollary. $\square$

6.1 Ordinary invariant subspaces and compactness

For every $h \in \mathbb{Z}$, the subspace $L^2(X; K_{\geq h})$ is closed, nonzero, proper, and S -invariant. It is not invariant under V : the image of the constant field e_h has its nonzero coordinate at $h - 1$ almost everywhere. Thus the construction separates ordinary invariance from hyperinvariance in an explicit way.

Each fiber S_x is compact. Indeed $a_k^\pm \geq c(k + 1)$ implies $\beta_j(x) \rightarrow 0$ as $|j| \rightarrow \infty$, and cutting off all but finitely many shift weights approximates S_x in operator norm. The operator S on H_0 is not compact. There is an infinite orthonormal sequence (φ_n) in $L^2(X)$, since the N_m disjoint residue classes give N_m orthogonal indicators for every m . The orthonormal vectors $\varphi_n \otimes e_0$ have images

$$S(\varphi_n \otimes e_0) = e^{-c} \varphi_n \otimes e_1.$$

These images are orthogonal with a common positive norm and have no norm-convergent subsequence.

6.2 The multiplier algebra has infinite multiplicity

Let Q_0 be the projection onto $L^2(X) \otimes \mathbb{C}e_0$. This gives a concrete witness to the properness of the commutant: on $1 \otimes e_0$,

$$Q_0 S(1 \otimes e_0) = 0, \quad S Q_0(1 \otimes e_0) = e^{-c}(1 \otimes e_1),$$

so $Q_0 \notin \{S\}'$.

The same projection commutes with every scalar multiplier, but is not itself a scalar multiplier. If $Q_0 = M_g$, its action on $1 \otimes e_0$ would give $g = 1$ almost everywhere, whereas its action on $1 \otimes e_1$ is zero. The abelian self-adjoint algebra $\{M_g : g \in L^\infty(X)\}$ is therefore not maximal abelian in $\mathcal{B}(H_0)$. Arveson's maximal-abelian inclusion theorem [1, Theorem 3.3] does not apply to this subalgebra of the transitive commutant.

6.3 Atomic finite families and a diffuse joint algebra

We finish with a spectral property of the constructed weights. This subsection is independent of the transitivity proof.

Let $|S| = (S^*S)^{1/2}$. Identify H_0 with $L^2(X \times \mathbb{Z}, \mu \times \#)$, where $\#$ is counting measure, and let $Ue_j = e_{j+1}$ on every fiber. Then

$$S = U|S|, \quad (|S|f)(x, j) = \beta_j(x)f(x, j).$$

All weights are positive, so $|S|$ is injective with dense range and U is the unitary polar part of S . Define

$$A_n = U^{-n}|S|U^n \quad (n \in \mathbb{Z}); \quad (A_n f)(x, j) = \beta_{j+n}(x)f(x, j).$$

For a family of operators, W^* denotes the smallest strongly closed unital self-adjoint algebra containing that family. A nonzero projection in an abelian von Neumann algebra is minimal if it contains no smaller nonzero projection in that algebra. Such an algebra is *atomic* when its identity is the strong sum of its minimal projections, and *diffuse* when it has no minimal projection. We will show that $W^*(A_n : n \in F)$ is atomic for every finite $F \subset \mathbb{Z}$, whereas $W^*(A_n : n \in \mathbb{Z})$ is diffuse.

For any finite $F \subset \mathbb{Z}$ and fixed j , the tuple $(\beta_{j+n}(x))_{n \in F}$ depends on finitely many residues $T_m(x)$. Because the residues are nested, it depends on one residue $T_M(x)$ and takes finitely many values. Allowing the countably many values of j gives a countable range for the joint tuple on $X \times \mathbb{Z}$. Its nonempty level sets have positive measure: on every fixed coordinate slice they are finite unions of residue classes. These level sets give the minimal projections of the generated abelian algebra, and their sum is the identity. Thus $W^*(A_n : n \in F)$ is atomic. In particular, $|S|$ has a pure-point spectral resolution.

For the full family, the two equal central weights determine the coordinate slice. They are the only weights equal to e^{-c} , since $a_k^\pm \geq c(k+1) > c$ for every $k \geq 1$. At (x, j) the tuple indices n for which $\beta_{j+n}(x) = e^{-c}$ are therefore exactly $-j-1$ and $-j$. For every $r \in \mathbb{Z}$, the projection onto the slice $j = r$, denoted by P_r , belongs to $W^*(A_n : n \in \mathbb{Z})$: it is

$$\mathbf{1}_{\{e^{-c}\}}(A_{-r-1}) \mathbf{1}_{\{e^{-c}\}}(A_{-r}).$$

Next fix $m \geq 1$ and choose $k_m = \ell_{m-1} \in I_m$. On the slice $j = r$, the multiplier A_{k_m-r} supplies $\beta_{k_m}(x)$. For $0 \leq t < N_m$, put

$$\lambda_{m,t} = \exp[-c(k_m + 1) - J_m - Dt].$$

Equation (17) shows that the multiplication projection of the residue rectangle $\{T_m = t\} \times \{r\}$ is

$$P_r \mathbf{1}_{\{\lambda_{m,t}\}}(A_{k_m-r}).$$

Thus every residue-class projection on every coordinate slice belongs to the generated algebra. Such rectangles generate the product sigma-algebra. Conversely all A_n are multiplication operators, so, modulo completion and null sets,

$$W^*(A_n : n \in \mathbb{Z}) = L^\infty(X \times \mathbb{Z}, \mu \times \#)$$

in its multiplication representation. Haar measure on $\mathbb{Z}_2$ has no atoms, and consequently this algebra is diffuse.

The pure-point assertion does not assert discreteness of the spectrum or compactness of S . Nor is this full multiplier algebra the scalar base algebra used in the transitivity proof. In fact $|S|$ does not commute with S , because the adjacent weights $\beta_0 = e^{-c}$ and $\beta_1 \leq e^{-2c}$ differ. The polar-conjugate algebra is therefore not contained in $\{S\}'$.

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